

March 15, 2025

Faisal D'Souza, NCO
Office of Science and Technology Policy
Executive Office of the President
2415 Eisenhower Avenue Alexandria, VA 22314

Submitted by email to [REDACTED]

Re: Request for Information (RFI) on the Development of an Artificial Intelligence (AI) Action Plan ("Plan")

About Dreadnode

Dreadnode is at the forefront of offensive AI Research and Development (R&D), building AI systems that demonstrate parity with, or exceed, human capabilities in the offensive security domain.

Dreadnode was founded in 2023 by Will Pearce, who built the AI red teams at Microsoft and NVIDIA, and Nick Landers, an accomplished offensive security engineer, VP of Research, and author of the Dark Side Ops penetration testing and adversary simulation training. They bring diverse domain expertise and rare, hands-on experience from NVIDIA, Microsoft, Meta, Cohere, NetSPI among other leading innovators.

Dreadnode is committed to America's dominance in the AI race. We believe that AI will transform cybersecurity by introducing new vulnerabilities and enabling new offensive capabilities. Now is the time to be on the offense, and Dreadnode combines cybersecurity and AI knowledge to test models to their limits and leverage them for the national mission.

Overview

Artificial Intelligence (AI) is one of the most powerful tools of our time. Unfortunately, amidst the significant steps the United States has made to advance innovations in machine learning, the AI industry is consistently met with equal forces of fear-mongering and consequently, threats of heavy regulation. We believe this to be misguided—between cutting-edge AI models and the agentic systems built on top of them, America holds the key to strengthening its own cyber defense and retaining an offensive edge over its adversaries.

Leveraging AI To Protect America: Today's cyberspace threat landscape is at an inflection point. From adversarial nations to sophisticated criminal enterprises, American cyberspace is under continuous attack. Dreadnode advocates for harnessing the power of

AI to defend the nation. Our work emphasizes AI-enabled security solutions that can anticipate and automate national defense measures—particularly in cyberspace.

We suggest that this administration:

- Facilitate the testing and adoption of automated vulnerability discovery and remediation (AVDR) solutions.
- Consolidate the efforts of the Chief Digital and Artificial Intelligence Office (CDAO), the Defense Digital Service (DDS) the Defense Innovation Unit (DIU), the Strategic Capabilities Office (SCO) without losing the inherent agility of DDS, DIU, and SCO.
- Maintain and elevate the National Institute of Standards and Technology (NIST) Testing Risks of AI for National Security (TRAINS) Taskforce.
- Establish a Code Continuity Task Force.

Attacking AI to Test Its Limits: Dreadnode believes that today’s AI systems are only the **beginning**. As systems evolve, exploring their full range of capabilities will require more advanced testing. Our team understands that adversarial machine learning research and testing of AI systems is critical to exposing security gaps, identifying solutions, and optimizing AI integrations into our broader ecosystems.

We suggest that this administration:

- Rebrand the Artificial Intelligence Safety Institute (AISi) as NIST’S Exploration of AI Advancement (NEXA), to include a Scientific Advisory Board. This would enable NIST to lead interagency AI model evaluation and performance metrics, and position the U.S. as the global leader in scientific AI standards.
- Create an AI Red Teaming Hub within NEXA.

Leveraging AI to Protect America

Today's global threat landscape has tested our ability to strengthen U.S. national security posture. Between the rise of non-state and unattributed threat actors in combination with escalations in the cyber and space domains, Dreadnode recognizes the importance of understanding and harnessing the power of AI to defend the nation. Our work emphasizes AI-enabled security solutions that can anticipate and automate national defense measures—particularly in cyberspace.

Automated Vulnerability Discovery and Remediation (AVDR)

Dreadnode has had the opportunity to support DARPA's [Artificial Intelligence Cyber Challenge](#) (AIXCC), a two-year competition that seeks to find and fix vulnerabilities in open-source software, which undergirds the majority of U.S. critical infrastructure. This program is one example of how the U.S. government can successfully convene both public and private organizations to create and operationalize critical security solutions.

Policy Recommendations:

- Facilitate the transition of winning Cyber Reasoning Systems (CRSs) from DARPA's Artificial Intelligence Cyber Challenge (AIXCC) by ensuring the continued evaluation of these CRSs in software test beds until they are optimized for real-world implementation.
- Incentivize the existing agency members within the AIXCC Transition Working Group to securely test and report on the results of CRS performance within designated software test beds.
- Similar to the [Cyber Incident Reporting for Critical Infrastructure Act of 2022](#) (CIRCA), create a legislative framework for the private sector to adopt best vulnerability management practices within a reasonable timeline as determined by NIST, the Cybersecurity and Infrastructure Security Agency (CISA) and the Department of Energy. AVDR solutions will only strengthen U.S. national security posture and position USG to serve as a conduit for mass adoption of critical technological innovation.

AI-Enabled Military

In order to match the speed and scale of AI-enabled attack vectors targeting the nation, U.S. military forces will require the training and resources to leverage and understand AI in real time—including, but not limited to deterrence operations and proportionality strategies. As the DoD focuses more on AI-enabled warfighting capabilities through the [Replicator Initiative](#) and other programs, it is imperative that systems relied on by the warfighter are secure, nimble, and reliable.

This will require deep insight into how these systems perform under pressure and serve our warriors. At Dreadnode, we have built Strikes: an AI capability development environment to gather data on AI agent and model performance. Strikes traces each step undertaken by

the model or agent and makes clearly visible how permutations in structure and design influence outcomes and successes. Given this insight into how models can be tested and evaluated, our suggestions are as follows.

Policy Recommendations:

- Consolidate the efforts of the Chief Digital and Artificial Intelligence Office (CDAO), the Defense Digital Service (DDS) the Defense Innovation Unit (DIU), the Strategic Capabilities Office (SCO) without losing the inherent agility of DDS, DIU, and SCO.
 - CDAO is particularly effective as an amplifier of interagency military operations, which requires an internal structure that often prohibits agility. For this reason, it is imperative that DIU, SCO, and even the Defense Digital Service (DDS) maintain enough independence to operate nimbly and quickly produce technological solutions for the warfighter.
- Establish an AI Capabilities Council, integrating industry representatives from small businesses to private AI labs, to identify real-world machine learning solutions in support of U.S. military forces.
 - An AI Capabilities Council could iterate upon initiatives such as the AI Rapid Capabilities Cell (AI RCC).
- Maintain the TRAINs Taskforce to include or run in parallel with the National Science Foundation's [Planning Grants to Create Artificial Intelligence \(AI\)-Ready Test Beds](#) for the sake of transparency in resource sharing and AI evaluations.

Automations to Streamline Bureaucracy

Many inefficiencies in software implementation and integration efforts stem from basic fundamental operational and functional differences between programming languages. AI-enabled code translation efforts can facilitate software migration and cross-platform development.

Policy Recommendations

- Establish a Code Continuity Task Force: a centralized federal effort focused on automated code translation. This would expand upon existing but disparate efforts to modernize legacy software programs, i.e. [IBM's watsonx Code Assistant](#) and DARPA's [TRACTOR](#) program to automate translations from C and C++ code to Rust.
- Within said task force, DIU and DARPA are currently best situated to lead contracts with companies and pilot specific code translation tools. NIST collectively oversees the standardization and testing in close coordination with CDAO, which can operationalize tested solutions across the DoD.

Attacking AI to Find Its Limits

To assess the performance of AI models and AI requires sufficient testing including AI red teaming. At Dreadnode we have built unique knowledge and tooling to do just that. In our

platform Strikes, we have set up a digital AI laboratory in which AI models and Agents are exposed to specified tasks, their performance tracked and traced, and their results catalogued. The resulting insights allow builders to understand which changes to models and agents result in which changes in outcomes. This supports not only the iterative research process that is key to development, but also the assessments of risks and trade-offs.

Furthermore, Dreadnode has built an AI red-teaming platform called Crucible for users to challenge and test out the newest AI models. The platform, which is open to the public, has shown how users can figure out ways to break and improve AI systems and agents.

Based on these unique insights, Dreadnode believes that the private sector is best positioned to optimize AI resiliency through internal evaluation techniques and benchmark tests. The successful development of advanced research tooling to test AI models and agents is enough to ensure any subsequent transition into real-world environments without cumbersome oversight or regulation. Any centralized federal efforts should reflect this responsible stewardship of cutting-edge AI innovations to ensure America maintains its lead in the global AI race.

Rebranding the Artificial Intelligence Safety Institute (AISI): NEXA

- The Artificial Intelligence Safety Institute (AISI) renamed as NIST'S Exploration of AI Advancement (NEXA) would position NEXA to lead interagency AI model evaluation and performance metrics, strengthening America's role as the global leader in scientific AI standards without slowing down private sector development and implementations.
- Under NEXA, existing partnership efforts between AISI and the U.S. Department of Defense's (DoD) Chief Digital and Artificial Intelligence Office (CDAO) would be formalized to identify and collaborate with test and evaluation (T&E) partners.
 - Scale AI has already demonstrated the value-add of joining forces with AISI for model evaluation and CDAO for customized LLM benchmark tests geared towards DoD use cases.
 - This approach should be expanded to more areas of application with new standardized quality criteria for DoD use cases as well as specialized domain knowledge in high priority areas such as cybersecurity.
 - To account for the variance of scoring between testing and evaluation providers CDAO should also expand the list of partners. This is especially true in highly technical areas in which different scoring systems can lead to different outcomes. Including more T&E partners will serve to strengthen AI integrations.
- NIST's convening power is paramount, particularly through the existing AISI [Consortium](#) (AISIC) of over 200 member companies and organizations. If AISIC were to expand and rebrand as the NEXA Consortium, it would be able to take on a **Scientific Advisory Board**. This change would enable more interagency

involvement, accounting for the National Science Foundation's (NSF) notable absence from AISIC

- o NSF is uniquely embedded within the nation's academic ecosystem, with unparalleled access to the country's leading research institutions. This makes NSF the most powerful engine for scientific discovery and technological innovation.
- NEXA would allow more opportunity for an **AI Red Teaming Hub**, wherein participants across federal agencies like CISA, the intelligence community, and private AI labs could streamline information sharing through classified threat intelligence sharing channels.
 - o For example, NIST's [AI Test, Evaluation, Validation, and Verification](#) (TEVV) red teaming framework works closely with CISA to [ensure AI systems are fit for purpose](#).
 - o While NIST's TEVV framework focuses on public sector priorities and standards, and NIST's AISI SEAL hones in on industry-led, containerized innovation, both of these initiatives speak to the value of AI red teaming and model testing.
 - o The more these teams can securely interact with one another and share their findings, the easier it will be to optimize AI models and encourage solution adoption at the enterprise level.

Contact

To get in touch with the team for further discussion, please email [REDACTED]

